

Associations of proton pump inhibitors with susceptibility to influenza, pneumonia, and COVID-19: Evidence from a large population based cohort study

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Abstract

Adverse effects of proton pump inhibitors (PPIs) have raised wide concerns. The association of PPIs with influenza is unexplored, while that with pneumonia or COVID-19 remains controversial. Our study aims to evaluate whether PPI use increases the risks of these respiratory infections. The current study included 160,923 eligible participants at baseline who completed questionnaires on medication use, which included PPI or histamine-2 receptor antagonist (H2RA), from the UK Biobank. Cox proportional hazards regression and propensity score-matching analyses were used to estimate the hazard ratios (HRs) and 95% confidence intervals (CIs). Comparisons with H2RA users were tested. PPI use was associated with increased risks of developing influenza (HR 1.32, 95%CI 1.12-1.56) and pneumonia (hazard ratio [HR] 1.42, 95% confidence interval [CI] 1.26-1.59). In contrast, the risk of COVID-19 infection was not associated with regular PPI use (HR 1.08, 95%CI 0.99-1.17), while the risks of severe COVID-19 (HR 1.19, 95%CI 1.11-1.27) and mortality (HR 1.37, 95%CI 1.29-1.46) were increased. However, when compared with H2RA users, PPI users were associated with a higher risk of influenza (HR 1.74, 95%CI 1.19-2.54), but not pneumonia or COVID-19-related outcomes. In conclusion, PPI users are associated with increased risks of influenza, pneumonia, as well as COVID-19 severity and mortality compared to non-users, while the effects on pneumonia or COVID-19-related outcomes under PPI use were attenuated when compared to the use of H2RAs. Appropriate use of PPIs based on comprehensive evaluation is required.

eLife assessment

This **useful** study aims to quantify associations between regular use of proton-pump inhibitors (PPI) - defined as using PPI most days of the week during the last 4 weeks at one cross-section in time - with several respiratory outcomes (6 different outcomes) up to several years later in time. Weaknesses were identified in the design of the study, such as the measurement of the primary outcome and also the potential of bias which is inherent to the study design, which means the manuscript provides **incomplete** evidence.

1. Introduction

Proton pump inhibitors (PPIs), as one of the most commonly used medications worldwide, have been utilized for treating various conditions related to excessive gastric acid secretion [1]. In the United States, the prescription of PPIs has doubled from 1999 to 2012, and the number of people taking PPIs is still increasing due to their availability over the counter [2]. However, concerns are being raised regarding the long-term and inappropriate use of PPIs, which have been linked to a wide range of adverse conditions, including osteoporotic fractures, renal failure, and vitamin deficiencies [3–5].

PPI-induced hypochlorhydria and gastrointestinal residence of pathogens might increase the risk of respiratory infections [6]. Cohort studies in the United Kingdom and the United States reveal the risks of developing community- and hospital-acquired pneumonia are increased by approximately 100% and 30%, respectively [7, 8]. In contrast, a nested case-control study based on the UK General Practice Research Database indicates long-term PPI therapy is not associated with increased risk for community-acquired pneumonia [9], and a retrospective cohort study involving 593 265 patients in Canada demonstrates no increased risk in developing pneumonia among PPI recipients [10]. Recently, attention has also been paid to the susceptibility to Coronavirus Disease 2019 (COVID-19) caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Based on 53,130 participants in the United States, a dose-dependent increased risk of COVID-19 positivity among PPI users is found [11]. Another Danish study also indicate a marginally increased risk of COVID-19 positivity [12], whereas studies based on the UK Biobank and a Korean cohort indicate nonsignificant association [13, 14]. Meta-analyses on the associations between the use of PPI and SARS-CoV-2 infection also demonstrate conflicting results [12, 15].

To date, the association between PPI and influenza remains unknown. The current evidence referring to PPI, pneumonia and COVID-19 is controversial. Previous studies had several limitations. For instance, the study based on the UK General Practice Research Database did not adjust for potential confounding variables for PPI indications [16]. Direct comparisons with histamine-2 receptor antagonist (H2RA) users were not conducted in previous studies to minimize confounding by indication. In addition, the findings by Almario *et al.* were based on patients with gastrointestinal symptoms, rather than the general population [11]. The previous UK Biobank study merely included participants tested for COVID-19 from March to June 2020 [13].

By leveraging the large-scale cohort and updated data in the UK Biobank, we aim to evaluate the associations between the regular use of PPIs and the susceptibility to respiratory infections, including influenza, pneumonia, and COVID-19.

2. Methods

2.1 Study population

The detailed information on study design for the UK Biobank was described previously [17]. Invitations were sent to about 9.2 million people who were aged 40-69, had capacity to consent, registered with the National Health Service (NHS), and lived within 25 miles of one of the assessment centers [18]. The participants were free to withdraw at any time [17]. Over 0.5 million participants were recruited from 22 assessment centers in Scotland, England, and Wales (specific locations of assessment centers are available at: <https://biobank.ndph.ox.ac.uk/ukb/field.cgi?id=54>) from 2006 to 2010. Information such as touch screen questionnaire, interview, blood pressure, eye measurements, physical measurements and so on was collected in the assessment centers (detailed content of assessments is available at: <https://biobank.ndph.ox.ac.uk/ukb/refer.cgi?id=100241>). Written informed consent was acquired from each participant, and ethical approval was obtained from the North West Multi-Center Research Ethics Committee (approval number: 11/NW/0382, 16/NW/0274, and 21/NW/0157). The current study has been approved under the UK Biobank project 83339. In this study, 11,171 participants with missing PPI medication data and 56,907 participants with missing covariate data were excluded, and we further restricted the cohort to the participants with available primary-care data. Among them, 1,298 participants without follow-up had been removed after initial exclusion. For the evaluation of associations with influenza, pneumonia, and other respiratory infections, those only with self-reported records and diagnoses were further excluded. For the COVID-19 infection and COVID-19-related outcomes, we excluded participants whose COVID-19 testing data were unavailable or who had died before the COVID-19 pandemic (Figure 1).

2.2 Definition of exposure

The exposure of interest was regular use of PPIs. Verbal interviews were conducted by a trained nurse if participants answered that they were regularly taking prescribed or over-the-counter medication on the touchscreen, in which ‘regular’ was defined as most days of the week for the last 4 weeks, and information on specific types of medications was further recorded, while no response to the question on the interview was considered missing data for PPI use. Short-term medications, for example, a 1-week course of antibiotics, were not recorded in the interview. Types of PPIs available in the UK Biobank included omeprazole, lansoprazole, esomeprazole, rabeprazole, and pantoprazole. The regular use of H2RAs was also defined by the above process. When comparing PPI users with H2RA users, participants who took both medications were excluded. Information on dose or duration of acid suppressant use was not available in the UK Biobank.

2.3 Definition of outcomes

The primary outcomes of interest were influenza, pneumonia, COVID-19 infections (Supplementary Table S1). Briefly, the first reported occurrences of respiratory system-related conditions within primary care data, the International Classification of Diseases (ICD)-10 codes were categorized by the UK Biobank (<https://biobank.ndph.ox.ac.uk/showcase/label.cgi?id=2410>). Influenza included those caused by identified influenza virus (J09-J10) and virus not identified (J11). Pneumonia was defined as that caused by viruses (J12), bacteria (J13-15), and other infectious organisms (J16-18).

COVID-19-related data in the UK Biobank (available from January 2020 to September 2021) based on follow-up of the participants was used [19]. COVID-19 infection mainly included information on positive COVID-19 tests, and patients with inpatient diagnoses or mortality due to COVID-19

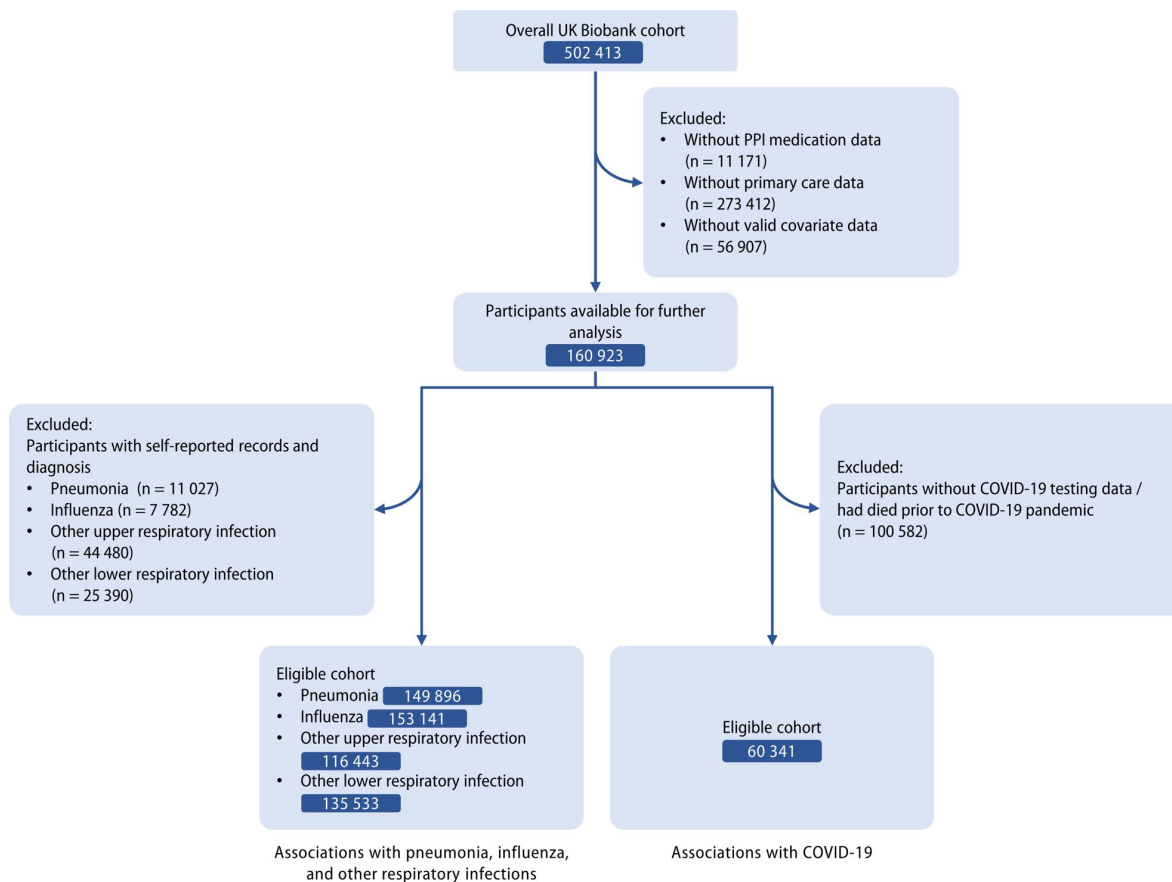


Figure 1.

Flow diagram of eligible participants selection.

were also regarded as having COVID-19 infection.

The secondary outcomes included other upper or lower respiratory infections, COVID-19 mortality, and COVID-19 severity. The definition of other upper respiratory infections contained acute nasopharyngitis, sinusitis, pharyngitis, tonsillitis, laryngitis, tracheitis, obstructive laryngitis, epiglottitis, or upper respiratory infections of multiple and unspecified sites (J00-J06). Other lower respiratory infections included acute bronchitis, bronchiolitis, and other unspecified ones (J20-J22). Severe COVID-19 cases were defined as being hospitalized for COVID-19. COVID-19 mortality was defined as the underlying recorded cause of death due to COVID-19 (ICD-10 U07.1 and U07.2).

2.4 Assessment of covariates

The covariates used for adjustments in our study were identified by a directed acyclic graph (DCA, **Supplementary Figure 1**) based on existing literature and expert knowledge. Baseline data on sociodemographic information (age, sex, ethnicity), socioeconomic status (deprivation index), alcohol consumption, smoking status, fresh fruit intake, multivitamin use, and body mass index (BMI) were collected from the UK Biobank, while physical activity was assessed using the International Physical Activity Questionnaire-Short Form. Gastroesophageal reflux disease (GERD), peptic ulcers, and upper gastrointestinal bleeding, were included as they are main indications for the use of PPIs. The comorbidities (hypertension, type 2 diabetes, renal failure, myocardial infarction, stroke, chronic obstructive pulmonary disease [COPD], asthma) were examined using self-reported data and adjusted due to their impact on the risk of respiratory infections. Since PPI and H2RA have highly similar indications, the use of H2RA was also adjusted. Data on medication use including aspirin, non-aspirin non-steroidal anti-inflammatory drugs (NSAIDs, including ibuprofen), and cholesterol-lowering medications were extracted and adjusted. For influenza and COVID-related outcomes, vaccinations were additionally adjusted.

2.5 CYP2C19 genetic variants

PPIs are mainly cleared by *CYP2C19*, and therefore their metabolism and effects are affected by different variants of *CYP2C19*. Genotyped genetic variant data after quality control was available for UK Biobank participants based on the Affymetrix Axiom UKB array and the Affymetrix UKBiLEVE array [20]. According to the Clinical Pharmacogenetics Implementation Consortium Guideline for *CYP2C19* and Proton Pump Inhibitor Dosing [21], genotypic data of four *CYP2C19* variants, including *rs12248560* (*CYP2C19**17), *rs17884712* (*CYP2C19**9), *rs4986893* (*CYP2C19**3), and *rs4244285* (*CYP2C19**2), were utilized to divide PPI users into three subgroups: (1) *CYP2C19* rapid and ultrarapid metabolizers (carried 1 functional allele and 1 increased-function allele [*17]; or carried 2 increased-function alleles); (2) *CYP2C19* normal metabolizers (carried 2 functional alleles); (3) *CYP2C19* likely intermediate, intermediate and poor metabolizers (carried ≥ 1 alleles with no/decreased function [*2, *3, and *9]).

2.6 Statistical analysis

The baseline characteristics were demonstrated by percentages for categorical variables, and mean (standard deviation [SD]), or median (interquartile range [IQR]) for continuous variables according to the distribution of data after evaluating the data distribution.

Univariate and multivariate Cox proportional hazards regression models were utilized to assess the association between regular use of PPIs and the selected outcomes, and the results were presented as hazard ratios (HRs) with 95% confidence intervals (CIs). Multivariable model 1 included age and sex. Model 2 additionally contained other potential confounders selected *a priori*, including ethnicity, deprivation index, alcohol consumption, smoking, physical activity, BMI, fresh fruit intake, GERD, peptic ulcer, upper gastrointestinal bleeding, hypertension, type 2 diabetes, renal failure, myocardial infarction, stroke, COPD, asthma, aspirin, non-aspirin NSAIDs, H2RA, cholesterol-lowering medications, and multivitamin use. The reference group was participants

without regular use of PPIs. Schoenfeld residuals tests were used to evaluate the proportional hazards assumptions, while no violation of the assumption was detected. Person-years were calculated from the number of participants and the date from Jan 2020 (for COVID-19-related outcomes) or recruitment (for other respiratory outcomes) to outcome diagnosis, last follow-up (September 2021 for COVID-19 infection and related outcomes; December 2021 for other outcomes), or death, whichever came first. Stratified analyses according to population characteristics, types of PPIs, and *CYP2C19* metabolizers were performed using multivariable-adjusted models across subgroups of each stratifying variable, and the multiplicative interactions were evaluated using likelihood ratio tests

Quantitative bias analyses were performed to calculate e-values, which illustrates the strength of association between an unmeasured confounder and exposure or outcome, conditional on the measured covariates [22]. E-value is the smallest magnitude of risk estimates that an unmeasured confounder would need to have with the exposure and outcome to explain away an observed association [22]. The event-free probabilities were compared by Kaplan-Meier survival curves. In addition, we conducted additional analyses using multiple imputation by chained equations to include participants initially excluded due to missing ethnicity data using the “mice” package [23]. Moreover, propensity score-matching analysis was conducted. The same set of covariates was used to derive propensity scores, and the PPI users and non-users were matched with a ratio of 1:4 using the “MatchIT” package [24], which estimated the propensity scores in the background and matched observations based on the nearest neighbor method. The remaining imbalanced covariates (standardized mean difference ≥ 0.1) after propensity score matching were further adjusted by multivariate Cox regression models to calculate HRs and 95% CIs [25]. Furthermore, because PPI and H2RA share highly similar indications, we performed head-to-head comparisons between PPI and H2RA users to further minimize the protopathic bias [26, 27].

All statistical analyses were performed using R (version 4.2.0, <https://www.r-project.org/>). The significance level at $\alpha = 0.05$ with two tails was used. Risk estimates were reported with 95% CIs.

3. Results

3.1 Study population

A total of 160,923 individuals aged 38 to 71 years who passed the initial selection criteria in the UK Biobank were included in this study (Table 1). The median follow-up was 7.1 (interquartile range [IQR] 6.2–8.5) years. The mean age of the included participants was 56.5 years, and 53.0% of them were women. Evidently, regular PPI users were characterized by higher rates of GERD (32.4% vs. 2.7%), peptic ulcer (5.6% vs. 0.9%), and upper gastrointestinal bleeding (0.2% vs. 0.03%) compared to non-regular PPI users. Higher burdens of comorbidities, as well as increased use of aspirin, H2RA, and cholesterol-lowering medications, were also observed in regular PPI users.

3.2 Proton pump inhibitor use and influenza, pneumonia, and COVID-19 infection

Increased risks of developing influenza, pneumonia, and other respiratory infections were identified in regular users of PPIs compared with non-regular users, and the risk remained raised after adjustments (Table 2). A 32% increased risk of developing influenza (aHR 1.32, 95% CI 1.12–1.56, $P = 0.001$; e-value 1.97) was observed among regular PPI users. In addition, regular use of PPIs was associated with a 42% increased risk of developing pneumonia (fully adjusted HR [aHR] 1.42, 95% CI 1.26–1.59, $P < 0.001$; e-value 2.19). Regular PPI users had lower event-free probabilities for influenza and pneumonia compared to those of non-users (Supplementary Figure 2 A–B). The

Characteristics	Regular PPI use		Overall
	Yes	No	
Number of participants, n (%)	9 997 (6.2)	150 926 (93.8)	160 923 (100.0)
Age, years, mean (SD)	59.4 (7.4)	56.3 (8.2)	56.5 (8.1)
Sex, female, n (%)	5 533 (55.4)	79 709 (52.8)	85 242 (53.0)
Ethnicity, white, n (%)	9 571 (95.7)	144 295 (95.6)	153 866 (95.6)
Deprivation index, mean (SD)	-0.9 (3.3)	-1.4 (3.0)	-1.4 (3.0)
Alcohol consumption, n (%)			
Daily or almost daily	1 805 (18.1)	28 846 (20.5)	32 874 (20.4)
3 or 4 times a week	1 923 (19.2)	33 533 (23.8)	37 570 (23.4)
1 or 2 times a week	2 396 (24.0)	37 443 (26.6)	42 228 (26.2)
1 to 3 times a month	1 179 (11.8)	15 669 (11.1)	17 988 (11.2)
Special occasions only	1 522 (15.2)	14 965 (10.6)	17 616 (11.0)
Never	1 161 (11.6)	10 414 (7.4)	12 570 (7.8)
Smoking, n (%)			
Never smoker	4 572 (45.7)	82 998 (55.0)	87 570 (54.4)
Previous smoker	4 289 (42.9)	52 013 (34.5)	56 302 (35.0)
Current smoker	1 136 (11.4)	15 915 (10.6)	17 051 (10.6)
Physical activity, MET minutes/week, median (IQR)	1 525.5 (2 722.0)	1 815.0 (2 848.5)	1 794.0 (2 838.5)
Fresh fruit intake, pieces, mean (SD)	2.0 (2.6)	1.9 (2.6)	1.9 (2.6)
BMI, kg/m ² , mean (SD)	29.2 (5.1)	27.4 (4.7)	27.5 (4.8)
Indication of PPIs, n (%)			
GERD	3 235 (32.4)	4 015 (2.7)	7 250 (4.5)
Peptic ulcer	561 (5.6)	1 303 (0.9)	1 864 (1.2)
Upper gastrointestinal bleeding	18 (0.2)	38 (0.03)	56 (0.03)
Comorbidities, n (%)			
Hypertension	4 116 (41.2)	38 162 (25.3)	42 278 (26.3)
Type 2 diabetes	124 (1.2)	890 (0.6)	1 014 (0.6)
Renal failure	60 (0.6)	243 (0.2)	303 (0.2)
Myocardial infarction	331 (3.3)	1 632 (1.1)	1 963 (1.2)
Stroke	140 (1.4)	943 (0.6)	1 083 (0.7)
COPD	46 (0.5)	200 (0.1)	246 (0.2)
Asthma	841 (8.4)	8 471 (5.6)	9 312 (5.8)
Medication use, n (%)			
Aspirin	2 457 (24.6)	21 108 (14.0)	23 565 (14.6)
Non-aspirin NSAIDs	1 224 (12.2)	22 568 (15.0)	23 792 (14.8)
H2RA	297 (3.0)	2 956 (2.0)	3 253 (2.02)
Cholesterol lowering medications	1 537 (15.4)	9 241 (6.1)	10 778 (6.70)
Multivitamin use, n (%)	2 227 (22.3)	33 201 (22.0)	35 428 (22.0)

BMI: body mass index; COPD: chronic obstructive pulmonary disease; GERD: gastroesophageal reflux disease; H2RA: histamine 2 receptor antagonist; IQR: interquartile range; MET: metabolic equivalent of task; PPI: proton pump inhibitor; NSAIDs: non-steroidal anti-inflammatory drugs; SD: standard deviation.

Table 1.

Baseline characteristics of the included participants.

association of PPI use with COVID-19 positivity was further evaluated in our study. Initially, in the non-adjusted model, the susceptibility to COVID-19 positivity was observed with a 18% increase (HR 1.18, 95% CI 1.09-1.26, $P < 0.001$ for non-adjusted model; **Table 2**) in participants with regular use of PPIs. However, full adjustments for covariates rendered the association nonsignificant (aHR 1.08, 95% CI 0.99-1.17, $P = 0.101$; **Table 2**).

3.3 Proton pump inhibitor use and other respiratory infections, COVID-19 severity, and COVID-19 morality

For other upper and lower respiratory infections, the risks among regular PPI users were increased by 19% (aHR 1.19, 95% CI 1.11-1.27, $P < 0.001$; e-value 1.67) and 37% (aHR 1.37, 95% CI 1.29-1.46, $P < 0.001$; e-value 2.08), respectively. In contrast, the risks of developing severe COVID-19 (aHR 1.33, 95% CI 1.09-1.61, $P = 0.004$; e value 1.99) and mortality due to COVID-19 (aHR 1.46, 95% CI 1.05-2.03, $P = 0.024$; e value 2.03) were significantly increased among PPI users compared to those among PPI non-users (**Supplementary Table S2**). PPI users had lower event-free probabilities for COVID-19 severity and mortality, but not COVID-19 positivity compared to those of non-users (**Supplementary Figure 2 C-E**).

3.4 Subgroup analysis

Stratified analyses were performed in the fully adjusted models for the main outcomes. Overall, no significant evidence of interactions was observed in the subgroup analyses referring to influenza (all P for interaction > 0.05 , **Figure 2**). The subgroup analyses for other main outcomes were illustrated in **Figure 2** and **Supplementary Figure 3**.

Among different types of PPIs, regular omeprazole or lansoprazole users were correlated with greater risks of respiratory infections(**Supplementary Table S3**). The risks of influenza were significant among *CYP2C19* normal metabolizers, and the risk estimate increased among *CYP2C19* likely intermediate, intermediate and poor metabolizers, while more information and larger sample sizes on PPI subtypes are still needed to increase the statistical power (**Supplementary Table S4**). The risks of COVID-19 severity and COVID-19 mortality were higher among *CYP2C19* likely intermediate, intermediate and poor metabolizers (**Supplementary Table S5**). The risks of pneumonia were higher among *CYP2C19* rapid and ultrarapid metabolizers (**Supplementary Table S4**).

3.5 Analysis by multiple imputation and propensity score-matching

After imputation of missing data, we found that individuals with regular use of PPIs were associated with similarly increased trends in the risks of influenza, pneumonia, other upper respiratory infections, and other lower respiratory infections (all $P < 0.05$) (**Supplementary Table S6**). The associations with COVID-19 severity and mortality were also significant (all $P < 0.05$) (**Supplementary Table S7**).

Matching of 9,910 regular PPI users and 39,760 non-regular users (1:4 by propensity scores) was also conducted, and the baseline characteristics were much more similar (**Supplementary Table S8**). The participants regularly exposed to PPIs were observed with increased risks for influenza, pneumonia, other upper respiratory infections, and other lower respiratory infections (all $P < 0.05$) (**Supplementary Table S9**), which were consistent with the results from Cox hazard proportional regression models. The associations with COVID-19 severity and mortality were also significant (all $P < 0.05$) (**Supplementary Table S10**).

	Case/person- years	Non-adjusted model		Age/sex-adjusted model		Fully adjusted model*	
		HR (95% CI)	P	HR (95% CI)	P	HR (95% CI)	P
Influenza							
Non-regular PPI use	2 009/6 011	1.00 (reference)		1.00 (reference)		1.00 (reference)	
Regular PPI use	183/539	1.38 (1.19-1.62)	<0.001	1.49 (1.28-1.74)	<0.001	1.32 (1.12-1.56)	0.001
			1		1		
Pneumonia							
Non-regular PPI use	2 904/12 867	1.00 (reference)		1.00 (reference)		1.00 (reference)	
Regular PPI use	378/1 702	2.04 (1.83-2.27)	<0.001	1.74 (1.56-1.94)	<0.001	1.42 (1.26-1.59)	<0.001
			1		1		
COVID-19 positivity							
Non-regular PPI use	23 989/29 080	1.00 (reference)		1.00 (reference)		1.00 (reference)	
Regular PPI use	1 440/1 702	1.18 (1.09-1.26)	<0.001	1.07 (0.99-1.15)	0.058	1.08 (0.99-1.17)	0.101
		6)	1	5)		7)	
Other upper respiratory infections							
Non-regular PPI use	14 449/52 499	1.00 (reference)		1.00 (reference)		1.00 (reference)	
Regular PPI use	1 118/3 988	1.30 (1.22-1.38)	<0.001	1.31 (1.23-1.39)	<0.001	1.19 (1.11-1.27)	<0.001
		8)	1	9)	1	7)	
Other lower respiratory infections							
Non-regular PPI use	14 494/55 384	1.00 (reference)		1.00 (reference)		1.00 (reference)	
Regular PPI use	1 486/5 598	1.78 (1.67-1.88)	<0.001	1.65 (1.56-1.74)	<0.001	1.37 (1.29-1.46)	<0.001
		8)	1	4)	1	6)	

*Adjusted for age, sex, ethnicity, deprivation index, smoking, alcohol consumption, physical activity, fresh fruit intake, body mass index, any indication of PPIs (gastroesophageal reflux disease [GERD], peptic ulcer, upper gastrointestinal bleeding), comorbidities (hypertension, type 2 diabetes, renal failure, myocardial infarction, stroke, chronic obstructive pulmonary disease [COPD], asthma), medications (aspirin, non-aspirin non-steroidal anti-inflammatory drugs [NSAIDs, ibuprofen], histamine 2 receptor antagonists (H2RAs), cholesterol lowering medications), multivitamin use, and influenza vaccination (for influenza) or COVID-19 vaccination (for COVID-19-related outcomes).

CI: confidence interval; COVID-19: coronavirus disease 2019; HR: hazard ratio; PPI: proton pump inhibitor.

Table 2.

Associations of PPI use with the susceptibility to pneumonia, influenza, COVID-19 positivity, and other respiratory infections.

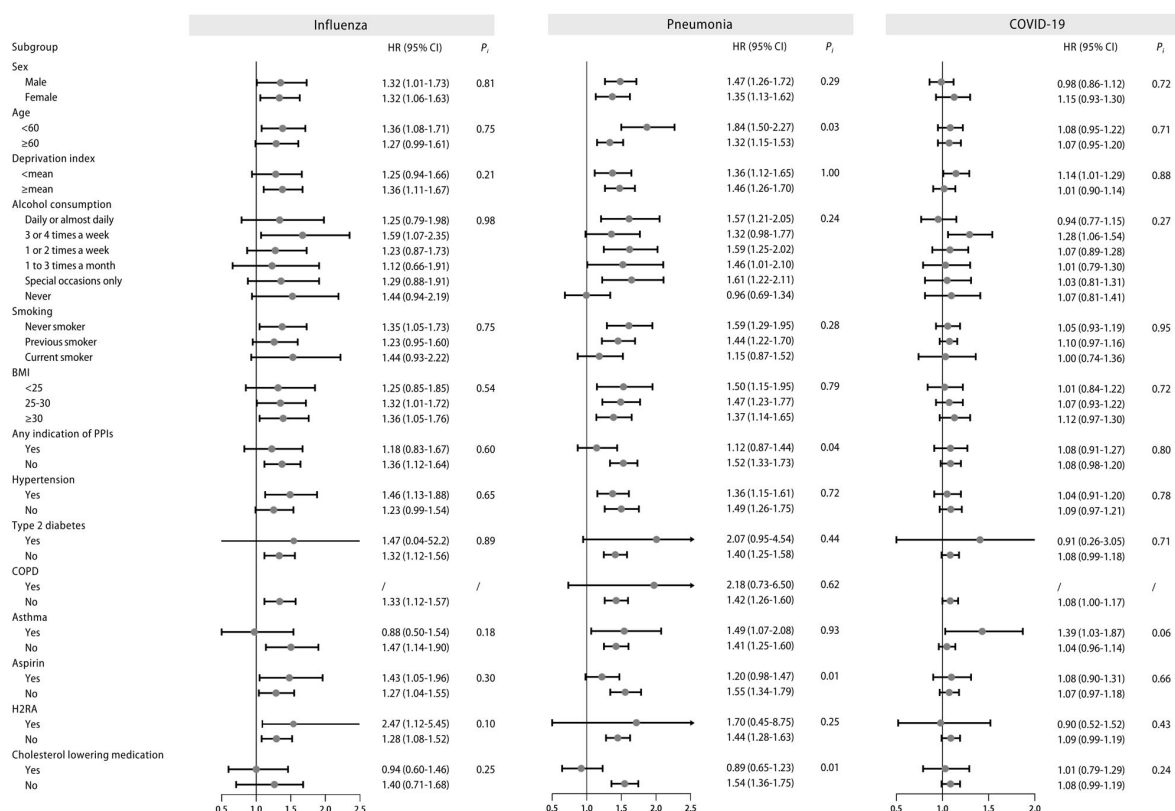


Figure 2.

Stratified analysis of regular proton pump inhibitor (PPI) users and the risk of influenza, pneumonia, and COVID-19 infection.

Effect estimates were based on age, sex, deprivation index, alcohol consumption, smoking, body mass index (BMI), indications of PPIs, hypertension, type 2 diabetes, chronic obstructive pulmonary disease (COPD), asthma, aspirin, histamine 2 receptor antagonist (H2RA), and cholesterol-lowering medication, using the fully adjusted model. CI: confidence interval; HR: hazard ratio; P_i : P value for interaction.

3.6 Comparisons with H2RA users

To further confirm the results and reduce the effect of confounding by indications, we evaluated the risk of respiratory infections compared to the use of H2RAs, which is a less potent acid-suppressant and contains indications similar to PPI. When compared to regular H2RA users, participants with regular use of PPIs were also associated with an increased risk of influenza (HR 1.74, 95% CI 1.19-2.54, $P = 0.004$; e-value 2.87), other upper respiratory infection (HR 1.28, 95% CI 1.07-1.54, $P = 0.008$; e-value 1.88), and other lower respiratory infection (HR 1.33, 95% CI 1.18-1.50, $P < 0.001$; e-value 1.99) (Table 3). However, the associations with pneumonia (HR 1.22, 95% CI 0.96-1.54, $P = 0.104$), COVID-19 infection (HR 1.04, 95% CI 0.87-1.26, $P = 0.629$), COVID-19 severity (HR 0.91, 95% CI 0.64-1.30, $P = 0.608$), or COVID-19 mortality (HR 0.83, 95% CI 0.45-1.56, $P = 0.745$) were not significant (Supplementary Table S11).

4. Discussion

In this large-scale, population-based cohort with updated information, we identify that the use of PPIs is associated with incident influenza. In contrast, analyses of pneumonia, COVID-19 infection and related outcomes, reveal attenuated effects after being compared with H2RA users. The association with influenza remains robust across different subgroups stratified by population characteristics and *CYP2C19* phenotypes.

The correlation between PPI use and the risk of influenza remains unexplored. For the past two decades, accumulating evidence indicates increased risks of pneumonia under the use of PPIs [7, 8, 28, 29], whereas others failed to show such associations [9, 10]. Conflicting findings also exist for studies referring to the risk of COVID-19 infection and related outcomes among PPI users, including several meta-analyses [11–15, 30–33]. Compared with existing studies, our study more comprehensively adjusts for a variety of critical covariates by utilizing the latest data from the UK Biobank. In addition, distinct from previous population-based studies, we compared the risks with H2RA users to further reduce protopathic and other unmeasured bias, since the users of acid suppressants, including PPIs and H2RAs, can have matched information on different characteristics, including indications. Although the risks of pneumonia were initially increased in Cox and propensity-score-matched analyses, direct comparison with H2RA users showed negative results, which indicates that previously observed associations could be due to unmeasured confounders.

Several proposed mechanisms can account for the association between the use of PPIs and respiratory tract infections. Since a low pH of gastric acid rapidly inactivates microorganisms, one critical issue is that reduced acidity induced by PPIs leads to the overgrowth of microorganisms, which can contribute to the development of infections in the respiratory tract through microaspiration [34]. Colonization and growth of pathogens under hypochlorhydria could increase the risk of respiratory infections. Although initial assessments indicated the use of PPIs might increase the risk of pneumonia, the head-to-head comparison with H2RAs yielded impacted effects. It could be due to the similar acid-suppressive effects of H2RAs and reduced sample size, which therefore warrants further investigations.

Concerns over protopathic bias due to non-specific and overlapping symptoms between influenza/pneumonia and acid-related diseases were raised [35]. Nevertheless, pneumonia and influenza often present with acute cough, and other concomitant symptoms distinct from acid-related diseases [36]. In contrast, patients with chronic cough are more commonly GERD-related [36]. The American College of Chest Physicians Clinical Practice Guidelines for Management of Reflux-Cough Syndrome also recommend against using PPI therapy alone for patients with chronic cough but without heartburn or regurgitation [37]. In our study, the use of PPIs is defined as

	Cases / Person-years	HR (95% CI)*	P
Influenza			
Regular H2RA use	32/102	1.00 (Reference)	
Regular PPI use	175/524	1.74 (1.19-2.54)	0.004
Pneumonia			
Regular H2RA use	86/385	1.00 (Reference)	
Regular PPI use	368/1653	1.22 (0.96-1.54)	0.104
COVID-19 positivity			
Regular H2RA use	425/506	1.00 (Reference)	
Regular PPI use	1 409/1 665	1.06 (0.90-1.24)	0.509
Other upper respiratory infection			
Regular H2RA use	146/522	1.00 (Reference)	
Regular PPI use	602/2099	1.28 (1.07-1.54)	0.008
Other lower respiratory infection			
Regular H2RA use	339/1350	1.00 (Reference)	
Regular PPI use	1438/5398	1.33 (1.18-1.50)	<0.001

CI: confidence interval; COVID-19: coronavirus disease 2019; H2RA: histamine-2 receptor antagonist; HR: hazard ratio; PPI: proton pump inhibitor.

*Adjusted for age, sex, ethnicity, deprivation index, smoking, alcohol consumption, physical activity, fresh fruit intake, body mass index, any indication of PPIs (gastroesophageal reflux disease [GERD], peptic ulcer, upper gastrointestinal bleeding), comorbidities (hypertension, type 2 diabetes, renal failure, myocardial infarction, stroke, chronic obstructive pulmonary disease [COPD], asthma), medications (aspirin, non-aspirin non-steroidal anti-inflammatory drugs [NSAIDs, ibuprofen], cholesterol lowering medications), multivitamin use, and influenza vaccination (for influenza) or COVID-19 vaccination (for COVID-19-related outcomes).

Table 3.

Comparisons of the risks of influenza, pneumonia, and COVID-19 between proton pump inhibitor (PPI) and histamine-2 receptor antagonist (H2RA) users.

taking the medication for most days of the week in the last 4 weeks, which is uncommon for acute cough. Although we cannot completely rule out protopathic bias, we have attempted to minimize it by adjusting for covariates including PPI indications, matching with propensity scores, and comparing with H2RA users.

For the risk of developing influenza, we analyzed the risks among different *CYP2C19* metabolizers for the first time, and further observed a significant increase among *CYP2C19* normal metabolizers compared to rapid and ultrarapid metabolizers. Although the risks of several outcomes, for example, influenza and pneumonia, for *CYP2C19* likely intermediate, intermediate and poor metabolizers are not statistically significant, they could be due to the limited sample size, and the risk estimates are higher compared to those among other types of metabolizers. Intriguingly, the risks of developing influenza and pneumonia are higher among *CYP2C19* rapid and ultrarapid metabolizers regularly taking PPIs compared to other types of metabolizers. Since our study exclusively involves participants with valid primary care data, such an increased risk might be to some extent contributed by the over-prescription or self-taking of PPIs under the undesired effects without following the proper strategy. Our findings are generally consistent with the assumption that slower metabolizers are associated with higher risks of adverse effects, while larger samples are needed to increase statistical power. Prescription of PPIs based on different *CYP2C19* metabolism subtypes is therefore important to reduce the adverse effects.

Our study has several strengths. First, our study utilizes the updated large-sample data from the UK Biobank and exclusively includes participants with valid records from primary care, which reduces the information bias. Second, a variety of covariates, especially for the indications of PPIs and the use of aspirin, which might contribute to indication or protopathic bias, have been adjusted to enhance the robustness of our results. Third, genotypic data of metabolic enzymes has been integrated into our study. Fourth, propensity score-matching analysis reduces a greater portion of bias, and analyses by propensity-score matching or multiple imputation derive consistent results. Fifth, adjustments for vaccination for COVID-19 and influenza has been performed in our study to reduce the confounding effects by vaccination. Furthermore, the comparison with participants using H2RA, a less potent acid suppressant with similar indications, further reduces the confounding by indication. The findings on the risk of influenza remain highly consistent across different strata and sensitivity analyses.

Limitations exist in our study. Information on dose and duration of PPI use, different types of pneumonia, and pneumococcus vaccination is currently not available from the UK Biobank. It was possible that PPI use was misclassified during the follow-up in the UK Biobank since PPI use was mainly assessed at baseline. However, no effect moderation was observed in subgroup analyses for PPI users with indications (more likely to regularly use PPIs for a long period) compared to those without indications, indicating the risks remained increased among long-term PPI users. Additionally, PPIs are indicated for *Helicobacter pylori* (*H. pylori*) eradication, whereas the UK Biobank does not contain adequate data. Thus, the indication for eradicating *H. pylori* is not adjusted in this study. The data on different PPI subtypes and COVID-19 infection and related outcomes are relatively small, which limits their power and still needs further investigation. Moreover, residual confounding might still exist due to the observational nature, while the quantitative bias analysis indicated that our result was robust to unmeasured confounding. Residual genotyping impacts of other enzymes, although affecting the metabolism to a lesser extent, might also exist. Although no significant differences were found between PPIs and H2RAs regarding the association with pneumonia and COVID-19-related outcomes, this could be due to the reduced sample size and power, which require larger cohorts to validate the effects. Furthermore, the current study is principally based on white British ancestry in the United Kingdom, and future exploration of other ancestries with comparisons is warranted.

Our findings could have essential implications for the prevention of respiratory infections and the de-prescribing of PPIs in clinical practice. Administration of PPIs can rapidly increase intragastric pH to higher than 6 after 2-4 hours [38]. Emerging evidence has revealed the inappropriate prescription of PPIs in both the primary and secondary care settings, and 33-67% of the patients did not take the drug according to their countries' criteria [39]. Similarly, the baseline characteristics of the included participants in our study demonstrate that approximately 60% of the regular PPI users do not have main indications. In addition, although influenza is usually self-limiting in healthy individuals, its risk of complications is significantly increased among pregnant women and people with immunosuppression or chronic diseases [40]. Therefore, comprehensive evaluation of PPI use is needed in clinical practice.

5. Conclusion

In conclusion, compared to non-users, people regularly taking PPIs are associated with increased susceptibility to influenza, pneumonia, as well as COVID-19 severity and mortality, while their association with pneumonia and COVID-19-related outcomes is attenuated after comparison with the use of H2RAs and remains to be further explored.

Data Availability

The UK Biobank data are available at <https://www.ukbiobank.ac.uk/> after approval of application.

Conflict of Interest Statement

The authors declared no conflict of interest.

Declaration of generative AI in scientific writing

This study does not involve generative AI in scientific writing.

Author Contributions

RJZ, YYM, LJZ, and DLL contributed equally to this work. HC, WHS, RJZ, YYM, and LJZ contributed to data extraction, data analyses, and manuscript drafting. DLL contributed to data interpretation and manuscript drafting. HHW, ZWZ, QY, and JWL contributed to manuscript drafting. HC, WHS, QC, and FWL contributed to study design, data interpretation, and final approval of the manuscript.

Acknowledgements

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Availability of Data and Materials

The UK Biobank data are available on application to the UK Biobank (www.ukbiobank.ac.uk .

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Abbreviations

- ACE2: angiotensin-converting enzyme 2
- BMI: body mass index
- CI: confidence interval
- COVID-19: Coronavirus Disease 2019
- DCA: directed acyclic graph
- GERD: gastroesophageal reflux disease
- H2RA: histamine-2 receptor antagonist
- HR: hazard ratio
- IQR: interquartile range
- NSAIDs: non-aspirin non-steroidal anti-inflammatory drugs
- PPI: proton pump inhibitor
- SARS-CoV-2: severe acute respiratory syndrome coronavirus 2
- SD: standard deviation.

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Reviewer #1 (Public Review):

Summary:

The current study aims to quantify associations between the regular use of proton-pump inhibitors (PPI) - defined as using PPI most days of the week during the last 4 weeks at one cross-section in time - with several respiratory outcomes up to several years later in time. There are 6 respiratory outcomes included: risk of influenza, pneumonia, COVID-19, other respiratory tract infections, as well as COVID-19 severity and mortality).

Strengths:

Several sensitivity analyses were performed, including i) estimation of the e-value to assess how strong unmeasured confounders should be to explain observed effects, ii) comparison with another drug with a similar indication to potentially reduce (but not eliminate) confounding by indication.

Weaknesses:

- (1) The main exposure of interest seems to be only measured at one time-point in time (at study enrollment) while patients are considered many years at risk afterwards without knowing their exposure status at the time of experiencing the outcome. As indicated by the authors, PPI are sometimes used for only short amounts of time. It seems biologically implausible that an infection was caused by using PPI for a few weeks many years ago.
- (2) Previous studies have shown that by focusing on prevalent users of drugs, one often induces several biases such as collider stratification bias, selection bias through depletion of susceptible, etc.
- (3) It seems Kaplan Meier curves are not adjusted for confounding through e.g. inverse probability weighting. As such the KM curves are currently not informative (or the authors need to make clearer that curves are actually adjusted for measured confounding).
- (4) Throughout the manuscript the authors seem to misuse the term multivariate (using one model with e.g. correlated error terms to assess multiple outcomes at once) when they seem to mean multivariable.
- (5) Given multiple outcomes are assessed there is a clear argument for accounting for multiple testing, which following the logic of the authors used in terms of claiming there is no association when results are not significant may change their conclusions. More high-level, the authors should avoid the pitfall of stating there is evidence of absence if there is only an absence of evidence in a better way (no statistically significant association doesn't mean no relationship exists).
- (6) While the authors claim that the quantitative bias analysis does show results are robust to unmeasured confounding, I would disagree with this. The e-values are around 2 and it is clearly not implausible that there are one or more unmeasured risk factors that together or alone would have such an effect size. Furthermore, if one would use the same (significance) criteria as used by the authors for determining whether an association exists, the required effect size for an unmeasured confounder to render effects 'statistically non-significant' would be even smaller.
- (7) Some patients are excluded due to the absence of follow-up, but it is unclear how that is determined. Is there potentially some selection bias underlying this where those who are less healthy stop participating in the UK biobank?
- (8) Given that the exposure is based on self-report how certain can we be that patients e.g. do know that their branded over-the-counter drugs are PPI (e.g. guardium tablets)? Some discussion around this potential issue is lacking.

(9) Details about the deprivation index are needed in the main text as this is a UK-specific variable that will be unfamiliar to most readers.

(10) It is unclear how variables were coded/incorporated from the main text. More details are required, e.g. was age included as a continuous variable and if so was non-linearity considered and how?

(11) The authors state that Schoenfeld residuals were tested, but don't report the test statistics. Could they please provide these, e.g. it would already be informative if they report that all p-values are above a certain value.

(12) The authors would ideally extend their discussion around unmeasured confounding, e.g. using the DAGs provided in <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7832226/>, in particular (but not limited to) around severity and not just presence/absence of comorbidities.

(13) The UK biobank is known to be highly selected for a range of genetic, behavioural, cardiovascular, demographic, and anthropometric traits. The potential problems this might create in terms of collider stratification bias - as highlighted here for example: <https://www.nature.com/articles/s41467-020-19478-2> - should be discussed in greater detail and also appreciated more when providing conclusions.

<https://doi.org/10.7554/eLife.94973.1.sa1>

Reviewer #2 (Public Review):

Summary:

Zeng et al investigate in an observational population-based cohort study whether the use of proton pump inhibitors (PPIs) is associated with an increased risk of several respiratory infections among which are influenza, pneumonia, and COVID-19. They conclude that compared to non-users, people regularly taking PPIs have increased susceptibility to influenza, pneumonia, as well as COVID-19 severity and mortality. By performing several different statistical analyses, they try to reduce bias as much as possible, to end up with robust estimates of the association.

Strengths:

The study comprehensively adjusts for a variety of critical covariates and by using different statistical analyses, including propensity-score-matched analyses and quantitative bias analysis, the estimates of the associations can be considered robust.

Weaknesses:

As it is an observational cohort study there still might be bias. Information on the dose or duration of acid suppressant use was not available, but might be of influence on the results. The outcome of interest was obtained from primary care data, suggesting that only infections as diagnosed by a physician are taken into account. Due to the self-limiting nature of the outcome, differences in health-seeking behavior might affect the results.

<https://doi.org/10.7554/eLife.94973.1.sa0>

Author Response

Reviewer #1 (Public Review):

Summary:

The current study aims to quantify associations between the regular use of proton-pump inhibitors (PPI) - defined as using PPI most days of the week during the last 4 weeks at one cross-section in time - with several respiratory outcomes up to several years later in time. There are 6 respiratory outcomes included: risk of influenza, pneumonia, COVID-19, other respiratory tract infections, as well as COVID-19 severity and mortality).

Strengths:

Several sensitivity analyses were performed, including i) estimation of the e-value to assess how strong unmeasured confounders should be to explain observed effects, ii) comparison with another drug with a similar indication to potentially reduce (but not eliminate) confounding by indication.

Thank you for pointing out the strengths of our article. We also sincerely thank the reviewer for raising several concerns and providing significant suggestions to improve our manuscript. We will revise our manuscript according to our provisional responses.

Weaknesses:

(1) The main exposure of interest seems to be only measured at one time-point in time (at study enrollment) while patients are considered many years at risk afterwards without knowing their exposure status at the time of experiencing the outcome. As indicated by the authors, PPI are sometimes used for only short amounts of time. It seems biologically implausible that an infection was caused by using PPI for a few weeks many years ago.

We agree with the reviewer, and this is one of the limitations of the UK Biobank data. We might identify potential long-term PPI users by defining the users that have certain indications, since they tend to regularly take PPI for a long period rather than only short amounts of time. We will evaluate the effect modification for the subgroup of potential long-term PPI users.

(2) Previous studies have shown that by focusing on prevalent users of drugs, one often induces several biases such as collider stratification bias, selection bias through depletion of susceptible, etc.

Due to the limitations of the data from the UK Biobank, including the lack of information on the initiation of medications and close follow-up, we can only use prevalent user design to evaluate the associations between PPI use and respiratory outcomes. We will further discuss it in the limitation section.

(3) It seems Kaplan Meier curves are not adjusted for confounding through e.g. inverse probability weighting. As such the KM curves are currently not informative (or the authors need to make clearer that curves are actually adjusted for measured confounding).

We will provide Kaplan Meier curves adjusted for confounding by inverse probability weighting according to the reviewer's suggestion.

(4) Throughout the manuscript the authors seem to misuse the term multivariate (using one model with e.g. correlated error terms to assess multiple outcomes at once) when they seem to mean multivariable.

We will correct the misused terms throughout the manuscript according to the reviewer's suggestions.

(5) Given multiple outcomes are assessed there is a clear argument for accounting for multiple testing, which following the logic of the authors used in terms of claiming there is no association when results are not significant may change their conclusions. More high-level, the authors should avoid the pitfall of stating there is evidence of absence if there is only an absence of evidence in a better way (no statistically significant association doesn't mean no relationship exists).

We will revise our interpretation of the results, especially for those without statistically significant associations based on the reviewer's advice.

(6) While the authors claim that the quantitative bias analysis does show results are robust to unmeasured confounding, I would disagree with this. The e-values are around 2 and it is clearly not implausible that there are one or more unmeasured risk factors that together or alone would have such an effect size. Furthermore, if one would use the same (significance) criteria as used by the authors for determining whether an association exists, the required effect size for an unmeasured confounder to render effects 'statistically non-significant' would be even smaller.

We agree with the reviewer that there might still exist one or more unmeasured risk factors that have effect sizes larger than 2. Therefore, we could not state that the results are robust to unmeasured confounding based on the current analysis, and this would be a limitation of our study. We will add the above information to the discussion section.

(7) Some patients are excluded due to the absence of follow-up, but it is unclear how that is determined. Is there potentially some selection bias underlying this where those who are less healthy stop participating in the UK biobank?

We will provide the details for the determination of absence of follow-up in the UK Biobank and illustrate whether it potentially induced selection bias.

(8) Given that the exposure is based on self-report how certain can we be that patients e.g. do know that their branded over-the-counter drugs are PPI (e.g. guardium tablets)? Some discussion around this potential issue is lacking.

In the data collection of the UK Biobank, the participants can enter the generic or trade name of the treatment on the touchscreen to match the medications they used. We will discuss this important issue in the discussion section.

(9) Details about the deprivation index are needed in the main text as this is a UK-specific variable that will be unfamiliar to most readers.

We will provide details about the deprivation index in the manuscript.

(10) It is unclear how variables were coded/incorporated from the main text. More details are required, e.g. was age included as a continuous variable and if so was non-linearity considered and how?

Age was included as a continuous variable. We will provide information on whether non-linearity was considered in our manuscript.

(11) *The authors state that Schoenfeld residuals were tested, but don't report the test statistics. Could they please provide these, e.g. it would already be informative if they report that all p-values are above a certain value.*

We will provide the test statistics for the Schoenfeld residuals.

(12) *The authors would ideally extend their discussion around unmeasured confounding, e.g. using the DAGs provided in <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7832226/>, in particular (but not limited to) around severity and not just presence/absence of comorbidities.*

We will use the DAGs provided by the article (PMC7832226) to extend our discussion around unmeasured confounding, especially the severity of comorbidities.

(13) *The UK biobank is known to be highly selected for a range of genetic, behavioural, cardiovascular, demographic, and anthropometric traits. The potential problems this might create in terms of collider stratification bias - as highlighted here for example: <https://www.nature.com/articles/s41467-020-19478-2> - should be discussed in greater detail and also appreciated more when providing conclusions.*

We agree with the reviewer that the highly selective nature of the UK Biobank might create collider stratification bias for the evaluation of COVID-19-related outcomes. We will further discuss this in detail and be cautious when generating conclusions.

Reviewer #2 (Public Review):

Summary:

Zeng et al investigate in an observational population-based cohort study whether the use of proton pump inhibitors (PPIs) is associated with an increased risk of several respiratory infections among which are influenza, pneumonia, and COVID-19. They conclude that compared to non-users, people regularly taking PPIs have increased susceptibility to influenza, pneumonia, as well as COVID-19 severity and mortality. By performing several different statistical analyses, they try to reduce bias as much as possible, to end up with robust estimates of the association.

Strengths:

The study comprehensively adjusts for a variety of critical covariates and by using different statistical analyses, including propensity-score-matched analyses and quantitative bias analysis, the estimates of the associations can be considered robust.

We thank the reviewer for demonstrating the strengths of our articles. We will further revise our manuscript according to the reviewer's suggestions.

Weaknesses:

As it is an observational cohort study there still might be bias. Information on the dose or duration of acid suppressant use was not available, but might be of influence on the results. The outcome of interest was obtained from primary care data, suggesting that only infections as diagnosed by a physician are taken into account. Due to the self-limiting nature of the outcome, differences in health-seeking behavior might affect the results.

We will try to adjust or provide discussions about the above factors, including the dose/duration of PPI use, outcome assessment, and health-seeking behavior.